TRAILBLAZER: generative multicellular model of biology

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Motivations (what questions did they aim to solve?)
Develop a model that can predict multicellular and patient-level responses to perturbations or therapeutic interventions while maintaining single-cell resolution.

Input Datasets
scRNA-seq data

Highlights
1. Mathematical foundation: By using a hyperspherical embedding, interventions and perturbations can operate via vector arithmetic, ensuring linearity.
2. All healthy cells are placed at the origin of the hypersphere, while different disease/treatment states are pointed outward on the hypersphere.
3. The mean positions for the healthy state and different disease/treatment states are calculated with all donors. For every type of intervention/perturbation, there is a displacement vector from the healthy center to it.
4. The mean of every state is simply the average of all latent vectors represented for all cells in that state.
5. Donor-dominance —> Perturbation-dominance: By forcing the displacement vector to align with the mechanism vector via cosine alignment, the perturbed states from all donors will be mapped to the same position on the hypersphere. In this way, it can ensure that the movement for a given perturbation/intervention is consistent across all donors, increasing generalizability.
6. A latent vector is produced by the induced set of attention blocks, which is a compressed form of a raw cell vector after “communicating with other cells”.
7. To capture the dynamics between cell interactions, instead of using a large O(N^2) matrix to represent individual cell-to-cell communication, the authors utilized a transformer-based encoder to learn from each of the cell and then redistribute all information back to each cell.
8. The mechanism segmentation network is pre-trained and frozen, serving as a library for calibrated mechanism changes. New unseen treatments can be added to this network by projection.

Future application/Implications

1. From a diseased state on the hypersphere to the healthy cell state, one specific and optimal addition/subtraction order between the mechanism vectors could be inferred, helping to find the most optimal strategy. (Evi: Fig. 4)
2. Can be applied in virtual drug screening, especially for the prediction of individual treatment responsiveness.
3. Latent shaping can improve the accuracy of zero-shot predictions.
4. The holdout strategy (zero-shot prediction) is commonly used for testing. No matter whether it is for unseen treatment, or held-out donors. The subject should be the one you want to test for. And it is a useful workflow to assess the performance of a computational model.